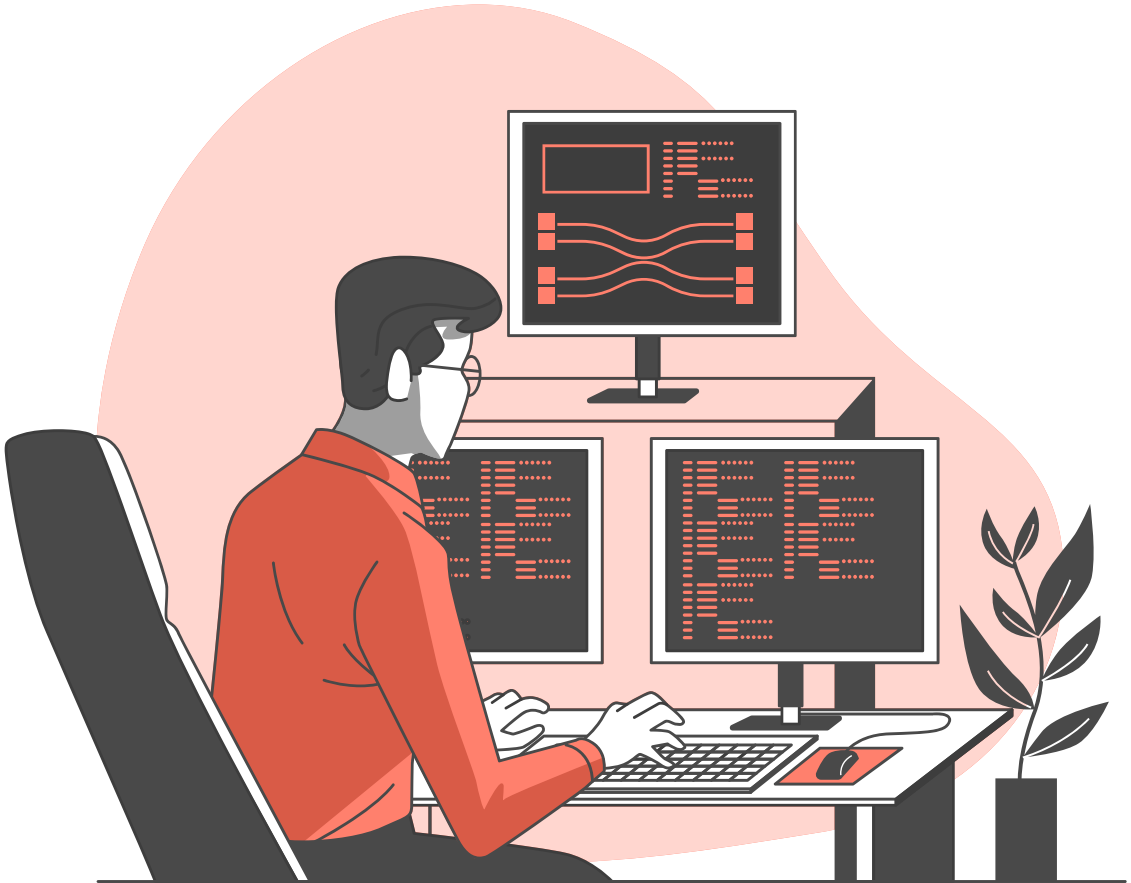

Session 0

Welcome to our lovely members!

Table of Contents

01 **Student activities**
What are they?
What they do?

02 **What is OSC?**
Who are we?
What is Open Source?
History of OSC

03 **OSC Hierarchy**
Who is the Board?
Who is the High Board?

04 **Backend Committee**
History
What do we do?
Committee Structure
Structure of each Session
Outline of each Level & Phases

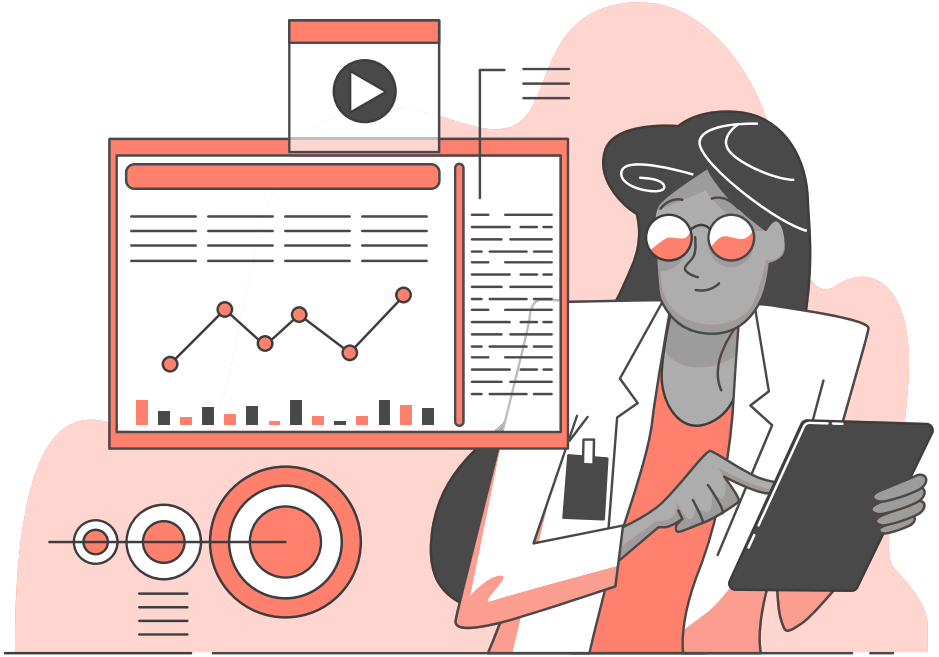

Table of Contents

05 **Backend Committee**
Punishment System
Rewarding System
Reporting System

06 **Backend**
What is Backend?
Why Backend?
How to be Backend Engineer?

07 **Why we are here?**
The value of what we do
How to get the most benefit?
What to expect from us?

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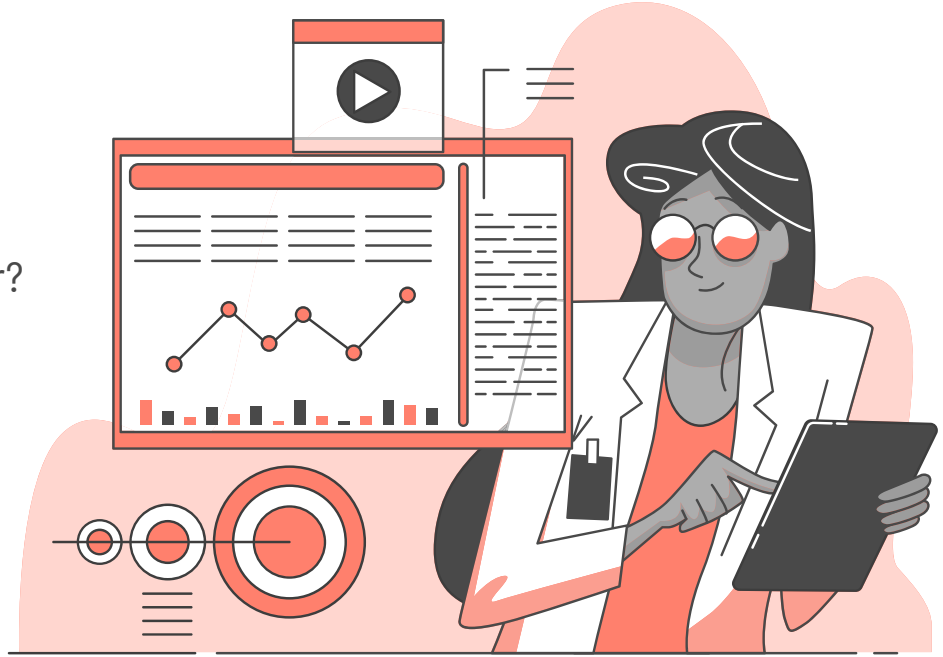


Student Activity

Student Activities

01 What Are They?

- What is student Activities?
- Why Do They Exist?

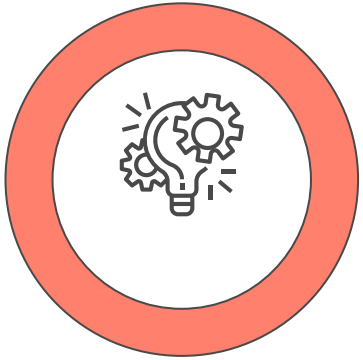
02 What They Do?

- Student Activities Roles
- Role

What Are Student Activities ?

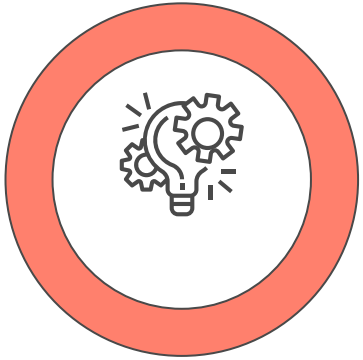
- Student activities are university-based communities or clubs

- Run by students to develop both Soft & Hard skills outside the classroom.



Why Do They Exist?

- To help students develop soft and technical skills.
- To bridge the gap between theory and practice.
- To create networks with like-minded people and mentors.
- To make university life more fun, active, and memorable.



Student Activities

01 What Are They?

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What Student Activities Do?

- They help students connect, learn, and grow together in the world of computer science and technology.

What Student Activities Do?

Through these activities, students get the chance to:

- build their skills
- share ideas
- work on exciting projects
- improving teamwork
- Communication
- leadership

Main Role

01

Soft-Skills

aiming to help students develop essential soft skills that prepare them for real-life challenges.
Through interactive workshops, teamwork activities, and practical sessions, participants will learn how to communicate effectively, manage their time, solve problems creatively, and work collaboratively.

02

Hard-Skills

focusing on developing strong technical skills, with a special emphasis on Technical fields.
aiming to help members understand Technical Concepts.
Through hands-on projects, coding sessions, and participants will gain practical experience in programming, and problem-solving

Main Role

03

Community Building

Bring students together to create a friendly, supportive space where everyone can grow, collaborate, and share knowledge.

Student Activities

01 What Are They?

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02 What They Do?

- Student Activities Roles
- Role

Guess the speaker?

OSC

OSC

01 Who are we?

02 What is Open Source?

03 History of OSC

04 Committees

05 Slogans

Who are we?

- OSC (Open Source Community) is a student activity that aims to spread the open-source mindset among students.

- We're a team of passionate learners who share knowledge, build projects, and grow together through collaboration and creativity.

What is Open Source?

- Open Source means that the source code or content is freely available for anyone to view, use, modify, and share.
- It's not just about software , it's a philosophy of transparency, collaboration, and community-driven growth.
- We believe learning should be shared , not locked away.

OSC History

OSC started 2013 by Adham Zahran who loves linux & open source very much.

Presidents across last 4 seasons

Nada → Hoda → Donia → Amr

Backend committee before Amr's season was web development which included frontend & backend.

What do you know about OSC’s history?

Committees

Technical	Non-Technical
Backend	Human Resources
Frontend	Public Relations
Flutter	UI/UX
Linux	Media
Science & Tech	Blender
Game	

Slogans

Don’t Reinvent The Wheel

زكاة العلم تعليمه

Get Ready to catch some Points

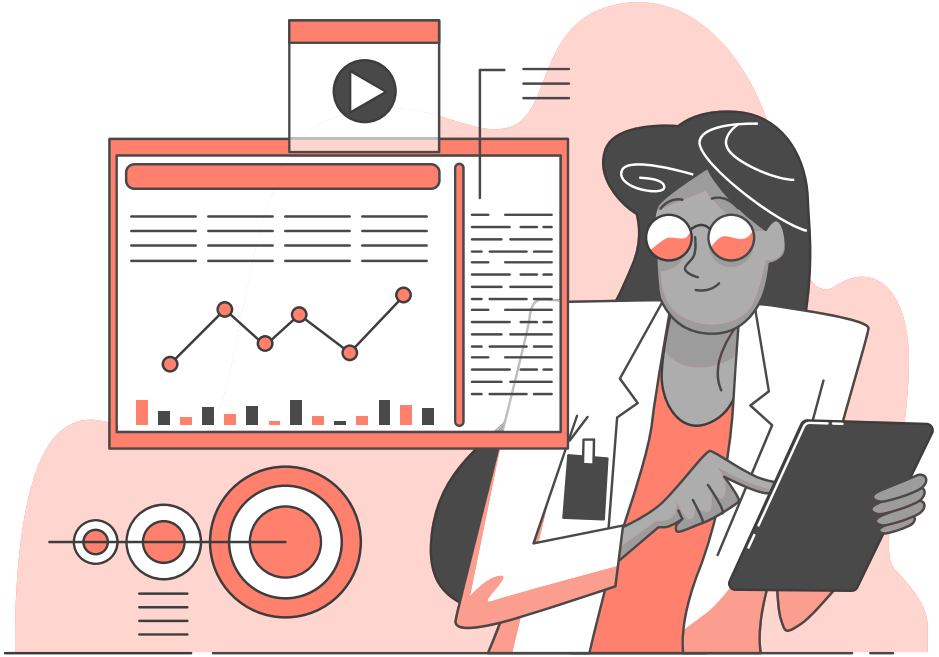
Table of Contents

01 ~~Student activities~~
~~What are they?~~
~~What they do?~~

02 ~~What is OSC?~~
~~Who are we?~~
~~What is Open Source?~~
~~History of OSC~~

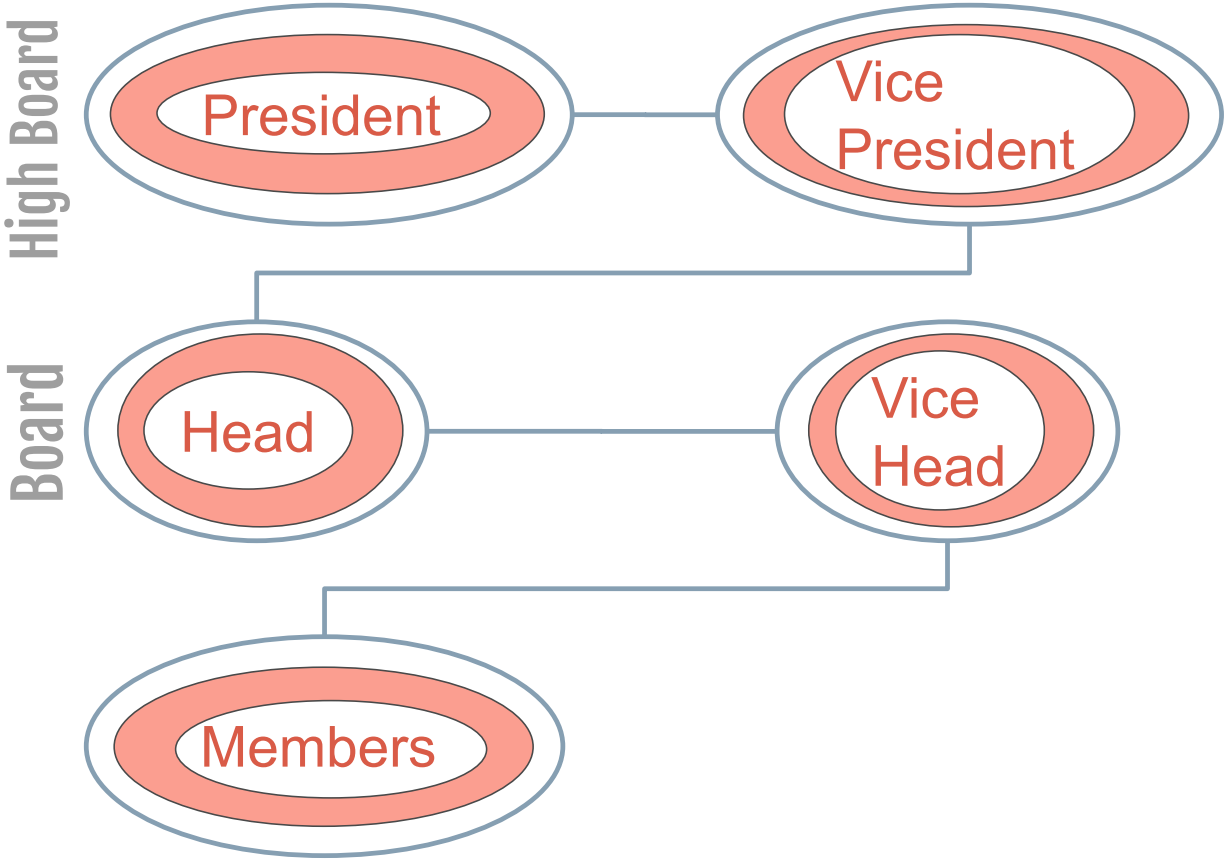
03 OSC Hierarchy
Who is the Board?
Who is the High Board?

04 Backend Committee
History
What do we do?
Committee Structure
Structure of each Session
Outline of each Level & Phases



OSC Hierarchy

OSC Hierarchy



High Board

Marwa Ahmed
President

Mohamed Bakr
Vice President
Back,Front,Flutter

Ghalia Hammed
Vice President
Linux, Science & Tech

Abdalla Osama
Vice President
Hr, Game ,Blender

Mariam Wael
Vice President
Pr,Media,UI/UX

Board

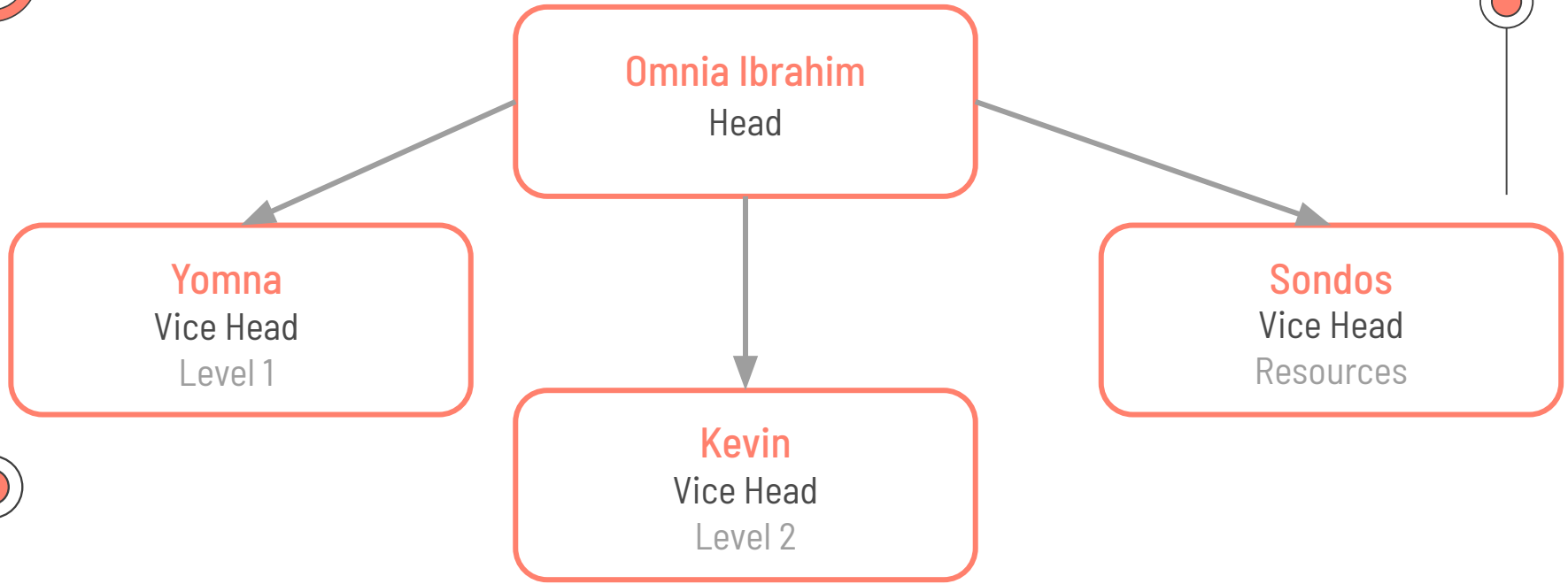

Omnia Ibrahim
Head
Yomna
Vice Head
Level 1
Kevin
Vice Head
Level 2
Sondos
Vice Head
Resources

Table of Contents

01 Student activities
- What are they?
- What they do?

02 What is OSC?
- Who are we?
- What is Open Source?
- History of OSC

03 OSC Hierarchy
- Who is the Board?
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- Structure of each Session
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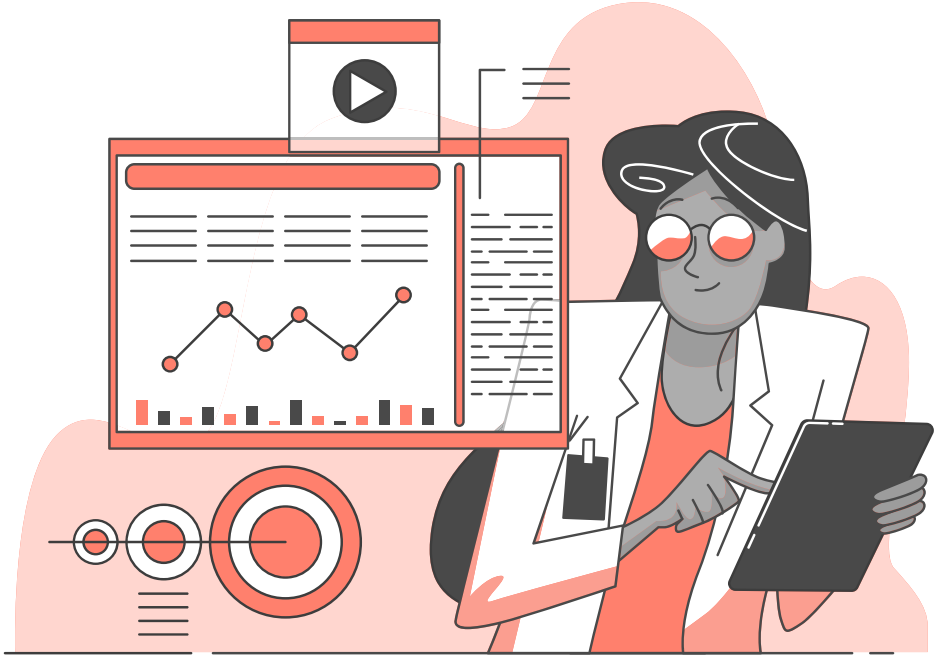


Guess the speaker?

Backend
Committee

Backend History

- The Backend Committee was established in Session 25 after the Web Committee was divided into two Committee : Frontend and Backend.

- The first Backend season was founded and led by Meefer (Mohamed Essam) and Amr Khaled.

What Do We Do ?

We aim to learn the fundamentals of backend development.
to build APIs, servers, and databases, and apply our skills through cumulative projects that combine everything we learn.

Committee Structure

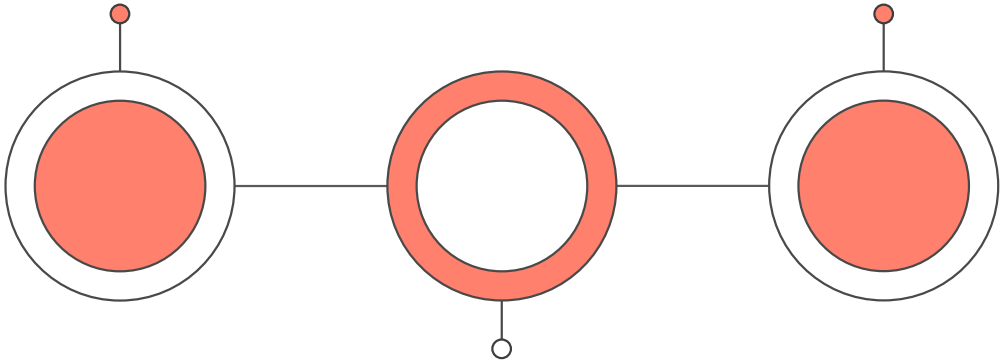
Level-2

Members from last Year Recruitment

Level-1

Members from This Year Recruitment

Sessions Structure



Levels Outlines & Content

Level-1

- JavaScript
- TypeScript
- Network
- Node JS
- Express JS
- SQL & NoSQL
- Authentication & Authorization

Level-2

- Nest JS
- SQL Database
- Caching
- File uploads
- Websockets webhooks
- System design & Architecture
- Microservices architecture
- Docker

Phases & Sessions TimeLine

- **5** sessions before the first freeze (before the practical exams)
- **4** sessions after the first freeze in the mid-year vacation
- **4** sessions before the 2nd midterm exams.
- **Project** sessions after the midterm exams

Table of Contents

01 Student activities
- What are they?
- What they do?

02 What is OSC?
- Who are we?
- What is Open Source?
- History of OSC

03 OSC Hierarchy
- Who is the Board?
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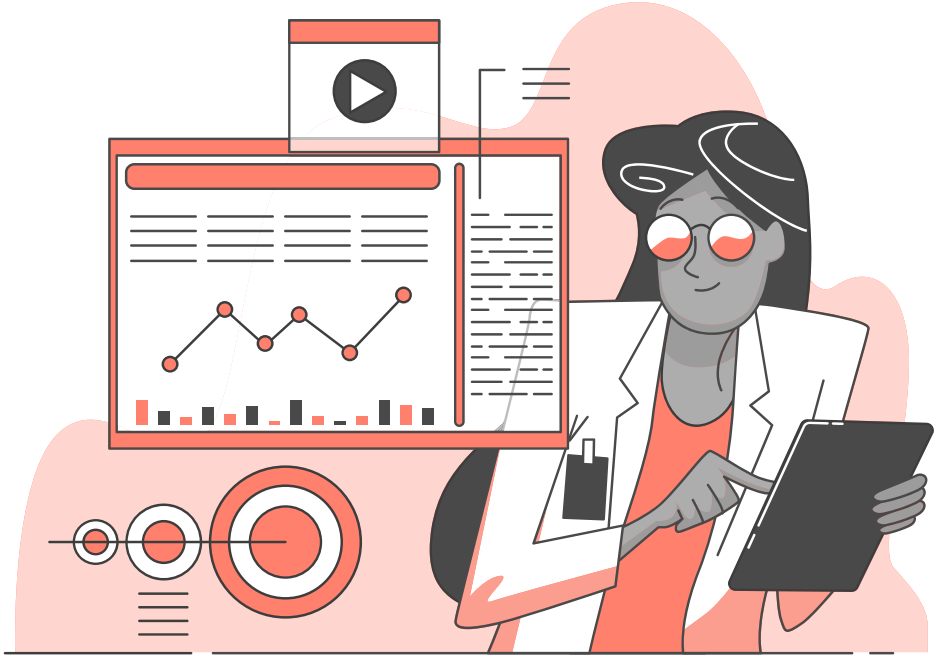

BREAK

CAUTION

There is a spy among us!!!!!!!

Guess Who is the HR

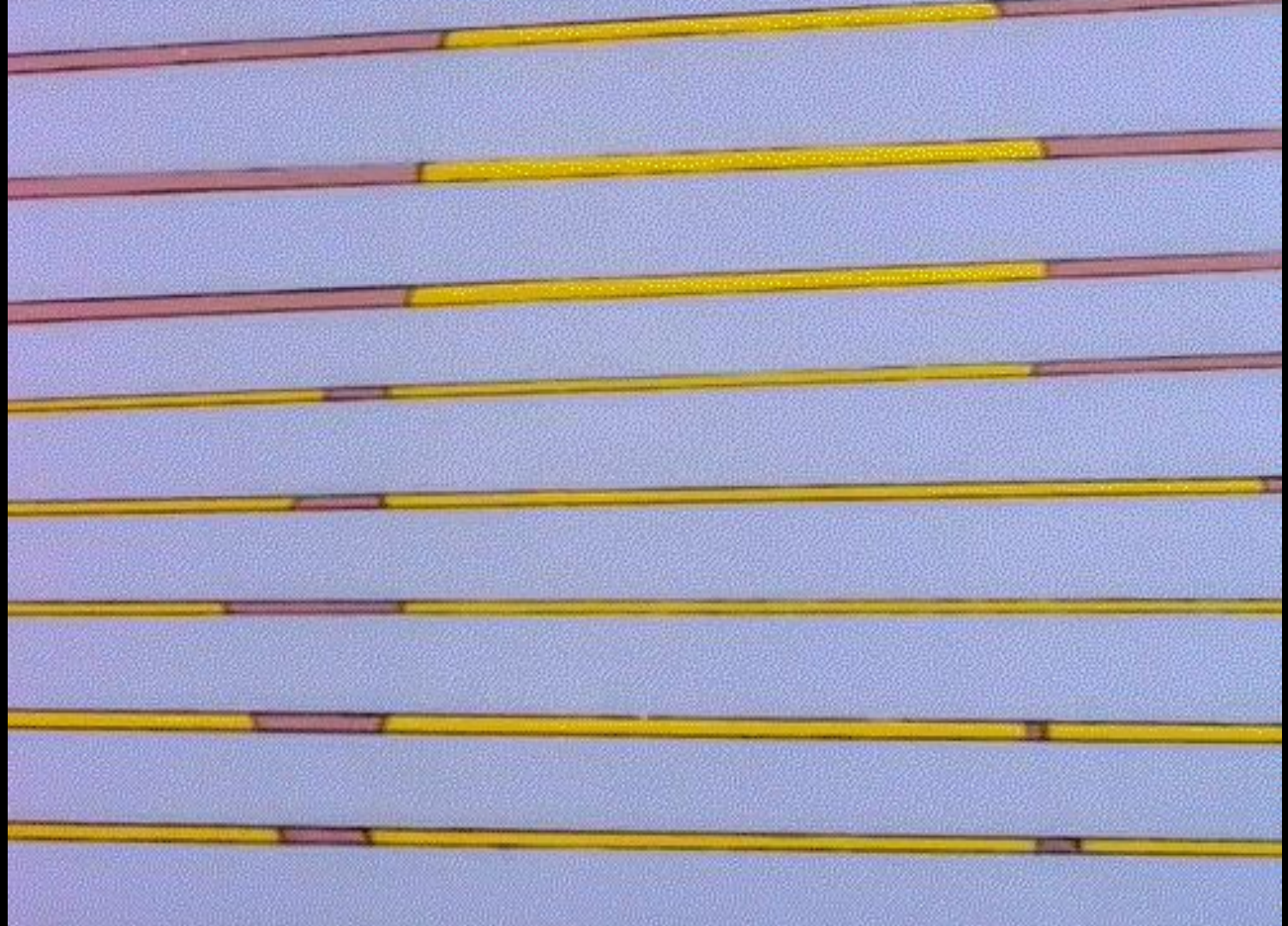

Table of Contents

05 **Backend Committee**
Punishment System
Rewarding System
Reporting System

06 **Backend**
What is Backend?
Why Backend?
How to be Backend Engineer?

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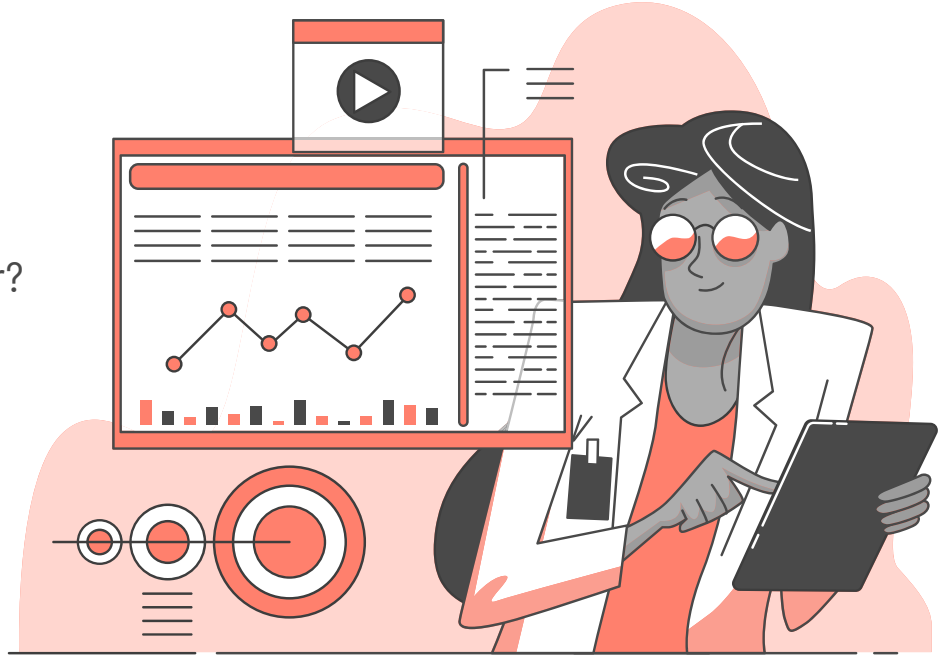
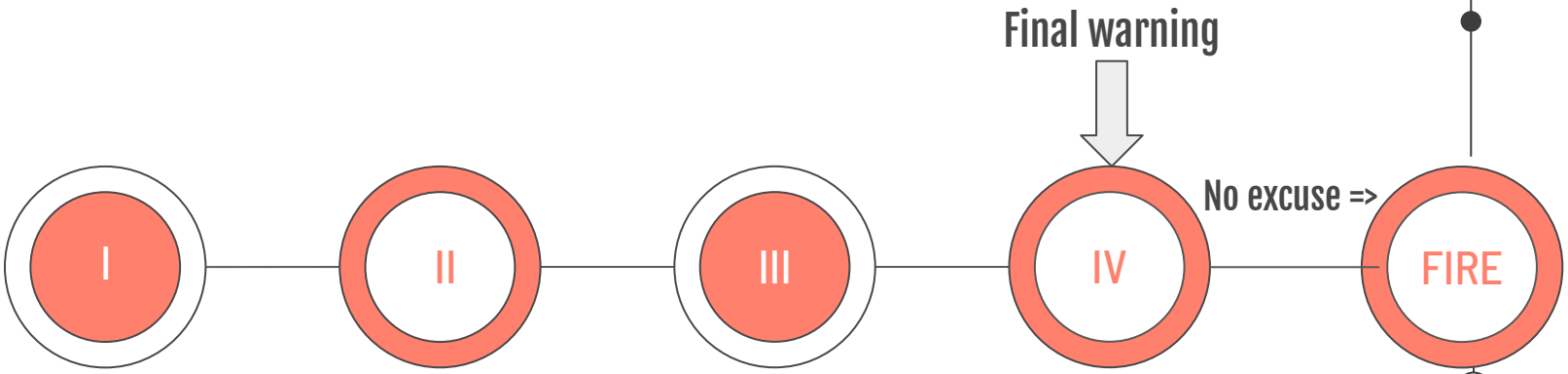


Punishment System



Not Submitting a Task Without Excuse => 1 warning
Not Attending a Session Without Excuse => 2 warnings

Warnings

Mohsens

- our activity's points system that rewards members for their effort and participation.
- Every time a member completes a task, attends a session, or contributes during activities, they earn points.

Rewarding System

Best of the Month

Based on the number of Mohsens received that month

Every month, the number of Mohsens is reset and we start recounting the Mohsens for each member

Best of the Season

Cumulative of all the months

The Mohsens counter doesn't reset every month ,points keep accumulating throughout the season.

Best of the Week

Based on task submission

Members who complete and submit their tasks on time, and with good quality; have a higher chance of being selected as the Best of the Week

Guess what is Badgician?

Badging System

It's a system designed to give you the space to compete, grow, and earn more Mohsens!

Here's how it works:

First of all when do you get a badge?

There will be a form with a clear description of each badge and what you need to do to earn it.
Once you complete the task for a specific badge, simply fill out the form.

After a quick review, your achievement will be announced on our Discord channel, and a certain number of Mohsens will be added to your score.

As you collect more Mohsens, your rank on the leaderboard will rise ,so keep learning, competing, and leveling up!

🏆 Badge Leaderboard

Top contributors and their achievements

#1	Kevin @kivenn 2 badges	★ ★
#2	Bakr @BakrDis 1 badge	★
#2	John doe @johnie 1 badge	★
#2	Mario @MarioN 1 badge	★

Last updated: 2025-10-29 22:24:31 UTC

Reporting System

- Imagine it as Layers

1. First layer is the HR (Non_Tech only)
2. Second Layer is Vices Heads
3. Third Layer is the Head of Your committee
4. Fourth Layer is the VP of Your committee
5. Fifth Layer is the President

Table of Contents

05 ~~Backend Committee~~
~~Punishment System~~
~~Rewarding System~~
~~Reporting System~~

06 Backend
What is Backend?
Why Backend?
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07 Why we are here?
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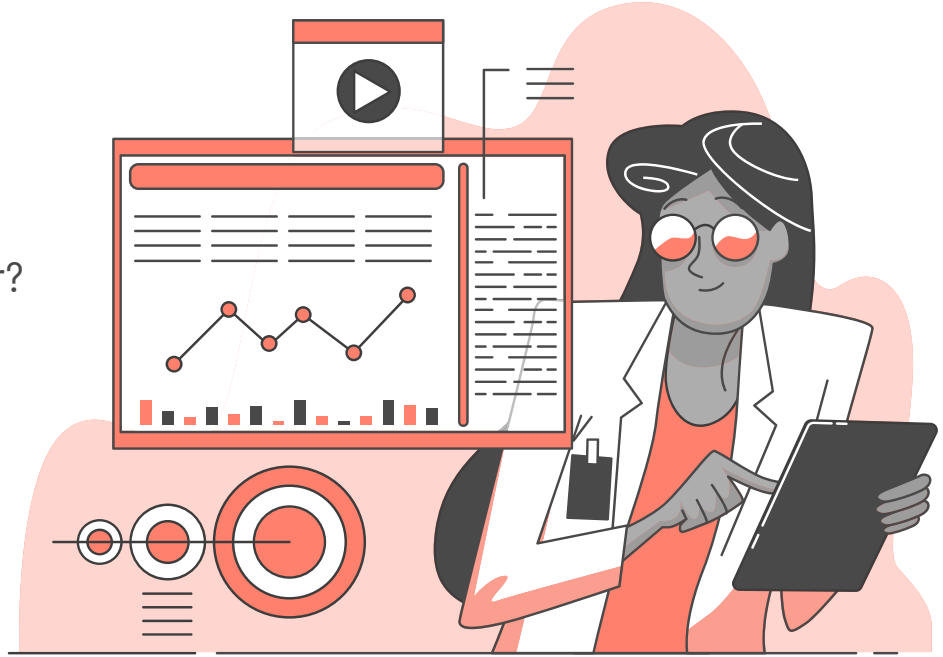


Guess the speaker?

What is Backend?

Why don’t we change things a bit and use another analogy rather than the good old classic restaurant one?

Any ideas?

The Human Brain Analogy

Backend :

Think of the backend as the brain of a human.
It manages all the inner functionalities , processing information, making decisions, and sending signals.

Frontend :

The frontend is like your face and expressions.
It shows what the brain wants to communicate, reacting to the signals sent behind the scenes.

Together, they create a seamless experience , the brain thinks, the face expresses.

The Building Analogy

Backend :

Think of the backend as the foundation and wiring of a building.

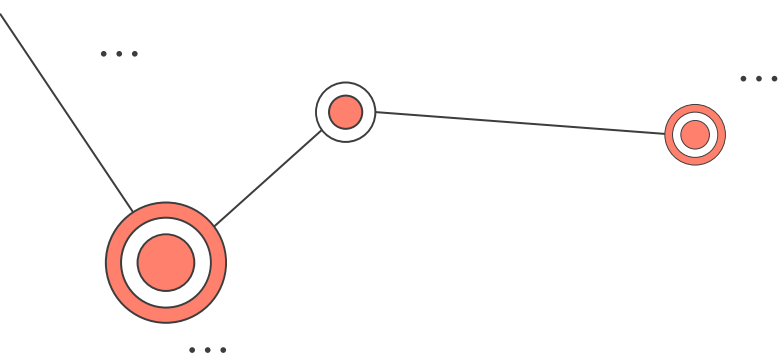
It supports everything behind the scenes — handling structure, stability, and the flow of power and data.

Frontend :

The frontend is like the interior design , the walls, colors, and furniture people actually see and interact with.

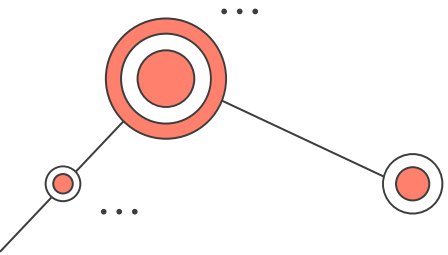
It takes what's built underneath and makes it visually appealing and user-friendly.

Together, they create a seamless experience , the backend builds stability, the frontend brings it to life.



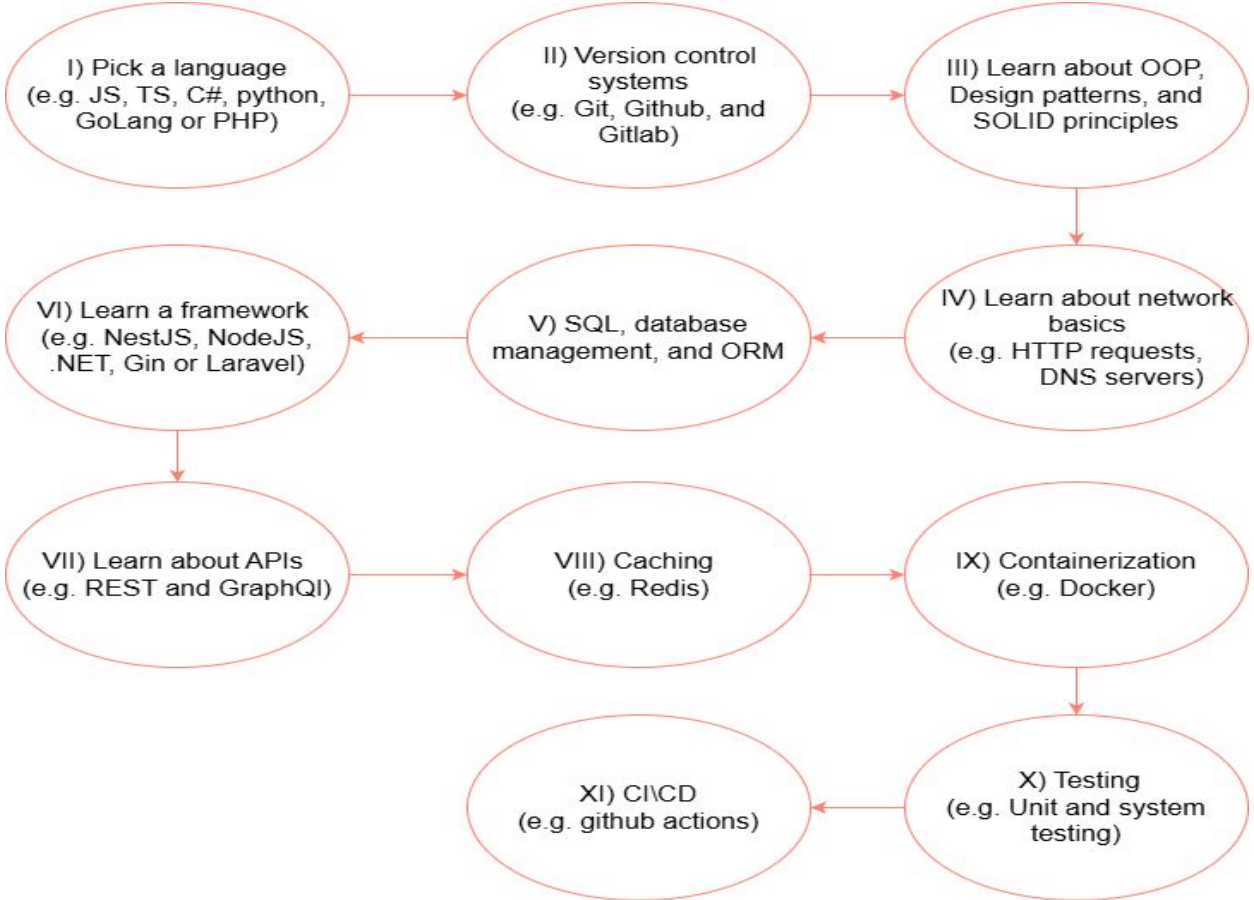
“The backend is where ideas become logic — it’s the invisible force that turns design into functionality.”

—Someone on reddit :)



How can I become a backend engineer?

Your average roadmap :)



You can also check this roadmap for reference.

What do you know about the backend job market requirements?

Software Engineer II Backend

talabat · Qesm El Maadi, Cairo, Egypt (On-site)

Easy Apply Save

Qualifications

- Empathetic and able to easily build relationships
- Good verbal and written communication skills
- Humble, curious, and open-minded with a growth mindset, thriving on continuous feedback
- Understanding of lean/agile values & principles and associated practices
- 3+ years of software engineering experience, preferably within Extreme Programming environment (TDD, Pair Programming, Simple Design, and Continuous Refactoring) with predominant languages being .NET, Golang, and C#
- Experience in testing with focus on automated end to end quality
- Experience in Distributed Systems Architecture (Microservices, Event-Driven, and RESTful Services)
- Experience working with relational and nonrelational databases
- Exposed to cloud platforms (e.g. AWS, Google Cloud, Azure)
- Experience building, maintaining and improving with Trunk Based Development and Continous Delivery

Google

Krakow, Poland

Experience: Mid-level

Posted: February 21, 2025

Golang Java Javascript Kotlin Nodejs Python

Backend

More jobs Salary Data

Requirements:

- Bachelor's degree or equivalent practical experience.
- 1 year of experience with software development in one or more programming languages (e.g., Python, C, C++, Java, JavaScript).
- 1 year of experience with data structures or algorithms.

Google

Zurich, Switzerland

Experience: Mid-level

Posted: November 4, 2024

Backend

More jobs Salary Data

Requirements:

- Bachelor's degree or equivalent practical experience.
- 1 year of experience with software development in one or more programming languages (e.g., Python, C, C++, Java, Javascript).
- 1 year of experience with data structures or algorithms.

Backend Engineer

Paymob · Cairo, Cairo, Egypt (Hybrid)

Apply Save

- Participate in code reviews to ensure high quality code delivery.
- Stay updated with the latest industry trends and technologies to continuously improve our systems.

Requirements

- Bachelor's degree in computer science, Software Engineering, or a related field.
- Minimum of 2-3 years of professional experience.
- Proven experience as a Backend Engineer or similar role.
- Strong proficiency in Python and its frameworks (e.g., Django, Flask).
- Experience with RESTful APIs and web services.
- Familiarity with database technologies (e.g., PostgreSQL, MySQL, MongoDB).
- Knowledge of version control systems (e.g., Git).
- Understanding of software development principles and design patterns.
- Experience with cloud services (e.g., AWS, Azure, Google Cloud) is a plus.

Table of Contents

05 ...

Backend Committee

Punishment System
Rewarding System
Reporting System

06 ...

Backend

What is Backend?
Why Backend?
How to be Backend Engineer?

07 ...

Why we are here?

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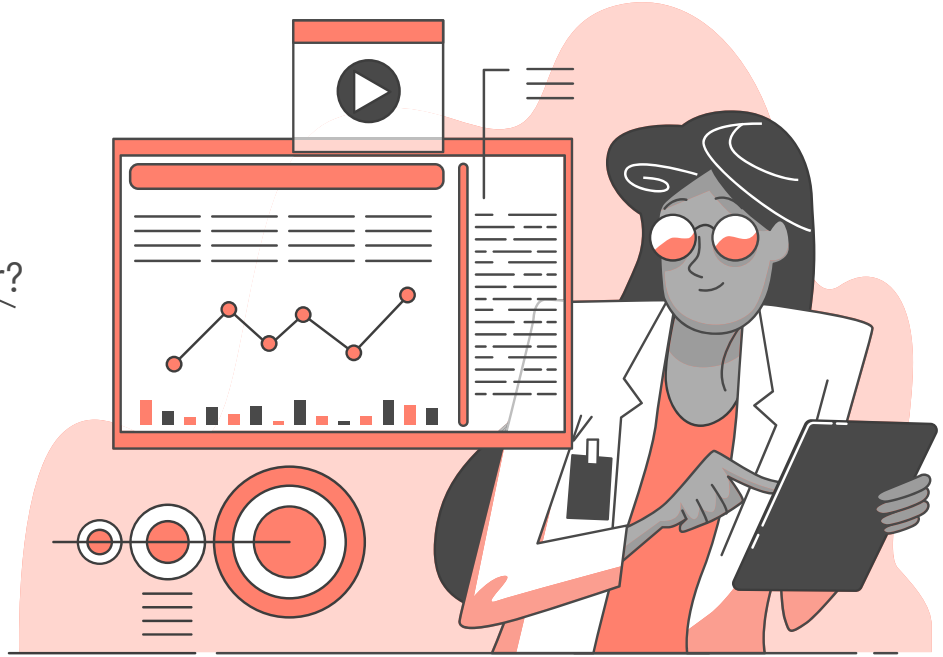


Guess the speaker?

The value of what we do

As we said before :"زكاة العلم تعليمه"

01 – **We give back**

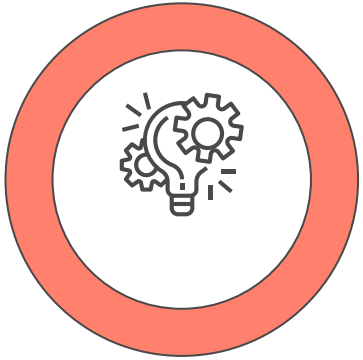
Everything we learn, we share with others — because our belief is simple

02 – **We learn by doing**

We don't just listen or attend sessions — we get our hands dirty and learn through real experience.

How to get the most benefit?

- ASK A LOT!!!!
- Share your opinion loudly!
- Read Documents.
- Minimize LLMs usage.
- Try things on your own first.
- Take notes — your future self will thank you.
- Don’t be shy — no one here knows everything.
- Help others when you can — it doubles your learning.
- Be curious, not perfect.
- Show up and stay consistent.
- Enjoy the process, not just the result.



What to Expect from us?

- Comprehensive explanation of the mentioned concepts.
- We'll do our best to explain every concept clearly and make learning fun.
- But we can't cover everything , you'll need to search, explore, and study on your own too.
- We'll guide you, but you're the one who'll grow through curiosity and effort.
- You'll get resources, materials, and support whenever you need them.
- Expect real projects, challenges, and discussions — not just slides.
- We'll be honest, approachable, and always here to help.
- And most importantly , we'll learn together, not as mentors and members, but as one community. ❤️

Table of Contents

05 Backend Committee
- Punishment System
- Rewarding System
- Reporting System

06 Backend
- What is Backend?
- Why Backend?
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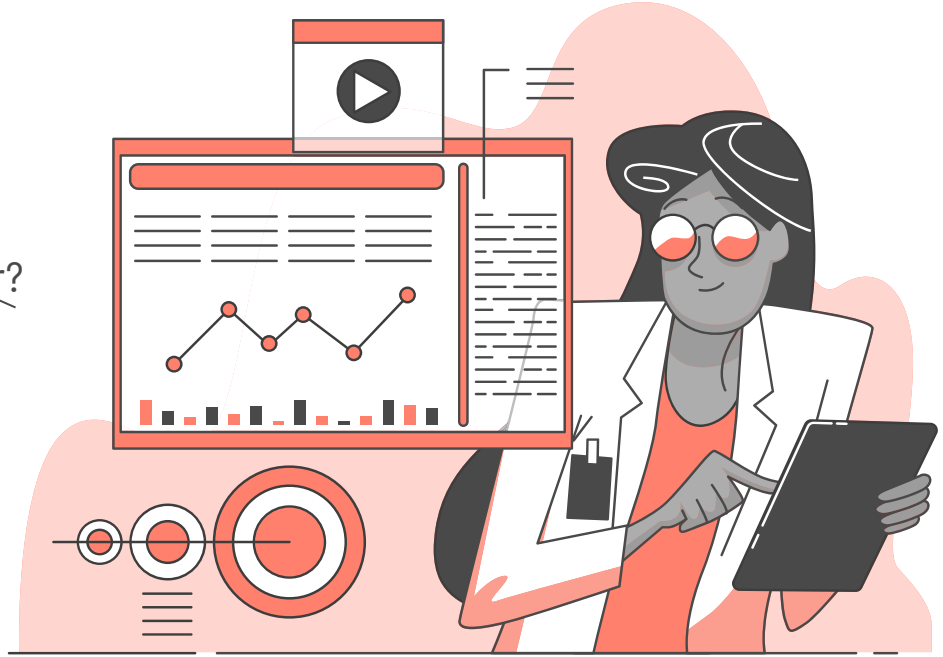

Surprise

Guess who is the Bonus Section Speaker?

Spin to Speak

Everyone gets a random number ,spin the wheel, and whoever's number appears gets the mic to introduce themselves.

PARTY TIME!!!!!

We will start out party with a small Game

Catch Mohsen!

- the instructor have a ball in his Hand
- displays a question
- throw the ball
- who catches the ball answers the question and gets mohsens
- The one catched the ball will be the one who throw it next time
- You can have the option to give it to someone (غلاسة)

READY

STEADY

GO!

- What is the main Role of a Student Activity?

Hint -> three cards

- What is Open Source?

- Say anything from OSC history

- Can you name the HIgh Board?

- What about the Board?

مش قد المقام يعني ولا ايه

- Tell us the Backend Committee history

- What was the Session Structure?

- Chose any system (punishment, Rewarding, Reporting)

And tell us about it

- Why backend?

- State any points from how to get the most benefit questions

PARTY TIME!!!!!

Thanks!

Do you have any questions?